

ONEDRAFT

CAD-AI — Aria Powered

Backup, Disaster Recovery & Business Continuity

Version 1.0

Status: Implementation Specification

System: OneDraft

Database: PostgreSQL

Artifact Storage: Object Storage

Execution: Runtime Orchestrator + Job & Worker Execution System

Recovery Model: Transactional, Versioned, Immutable and Reconstructable

Primary Objective: Preserve the ability to recover OneDraft project state, generated artifacts, operational services and system configuration following hardware failure, software failure, data corruption, security incident, infrastructure loss or regional service interruption.

1. PURPOSE

The OneDraft Backup, Disaster Recovery & Business Continuity subsystem establishes the mechanisms required to preserve and restore the operational integrity of the OneDraft platform.

The system must protect:

Project data

Project model state

Revision history

Geometry references

Compliance manifests

Validation results

Generated drawings

Document packages

Audit events

Identity and authorization state

Configuration

Infrastructure state

Operational metadata

Recovery must preserve the provenance chain established by the preceding OneDraft architecture.

The objective is not merely to restore files.

The objective is to restore a **known, internally consistent OneDraft system state**.

2. RECOVERY PRINCIPLE

OneDraft recovery shall operate according to the principle:

```
BACKUP
↓
VERIFY
↓
DETECT FAILURE
↓
ISOLATE
↓
RESTORE
↓
RECONCILE
↓
VALIDATE
↓
RESUME
```

A successful database restore without corresponding artifact recovery is not considered a complete recovery.

Likewise, restored artifacts without their associated model, revision and provenance state are not considered a complete recovery.

3. BUSINESS CONTINUITY OBJECTIVE

The platform shall maintain continuity of the following core capabilities:

```
AUTHENTICATION
ORGANIZATION ACCESS
PROJECT ACCESS
PROJECT MODEL
REVISION HISTORY
COMPLIANCE STATE
VALIDATION STATE
DRAWING GENERATION
DOCUMENT GENERATION
ARTIFACT RETRIEVAL
```

Non-critical derived services may be restored after the transactional project foundation.

4. RECOVERY DOMAINS

The recovery architecture shall distinguish:

DATABASE RECOVERY
ARTIFACT RECOVERY
APPLICATION RECOVERY
WORKER RECOVERY
QUEUE RECOVERY
CONFIGURATION RECOVERY
IDENTITY RECOVERY
SECRETS RECOVERY
OBSERVABILITY RECOVERY
INFRASTRUCTURE RECOVERY

Each recovery domain shall have an explicit restoration procedure.

5. SYSTEM OF RECOVERY RECORD

The authoritative recovery state is derived from:

PostgreSQL
+
Object Storage
+
Immutable Audit Records
+
Configuration Repository
+
Deployment Definitions
+
Secrets Recovery Mechanism

The recovery system must preserve relationships between these components.

6. RECOVERY UNIT

The fundamental recoverable OneDraft unit is:

Organization
↓
Project

↓
Model Version
↓
Revision
↓
Code Manifest
↓
Validation Run
↓
Drawing Set
↓
Document Package
↓
Acceptance

A recovery process must preserve these relationships.

7. DATABASE BACKUP

PostgreSQL shall be backed up using a combination of:

Continuous WAL archiving

Periodic full database backups

Logical backups where required

Verified restoration snapshots

The preferred production recovery mechanism shall support point-in-time recovery.

8. DATABASE POINT-IN-TIME RECOVERY

The production database shall support restoration to a specific timestamp.

Conceptually:

FULL BASE BACKUP
+
WAL ARCHIVE
=
POINT-IN-TIME RECOVERY

The target recovery timestamp shall be recorded during an incident.

The restored database must be validated before being promoted as authoritative.

9. DATABASE BACKUP FREQUENCY

The production database shall use:

Continuous WAL archival
Daily backup verification
Regular full database snapshots
Periodic restoration testing

Backup frequency may be increased as transaction volume increases.

10. DATABASE BACKUP RETENTION

The initial retention policy shall distinguish:

SHORT-TERM OPERATIONAL BACKUPS
MEDIUM-TERM RECOVERY BACKUPS
LONG-TERM ARCHIVAL BACKUPS

Retention periods shall be configurable through the administrative control plane.

Retention policy must not permit deletion of records that are required for contractual, regulatory, audit or provenance obligations.

11. BACKUP ENCRYPTION

Production backups shall be encrypted at rest.

Encryption shall apply to:

DATABASE BACKUPS
WAL ARCHIVES
OBJECT STORAGE BACKUPS
CONFIGURATION BACKUPS
RECOVERY EXPORTS

Encryption keys must not be stored alongside the backup material they protect.

12. BACKUP TRANSPORT SECURITY

Backup data transmitted between systems shall use encrypted transport.

Backup channels shall be isolated from ordinary application traffic where practical.

The backup system must authenticate both source and destination.

13. OBJECT STORAGE BACKUP

Generated OneDraft artifacts shall be protected independently from PostgreSQL.

Protected artifact classes include:

DRAWINGS
PDF DOCUMENTS
CAD FILES
BIM FILES
IMAGES
SURVEY DATA
SOURCE DOCUMENTS
CODE SOURCE ARTIFACTS
VALIDATION REPORTS
EXPORTS
ARCHIVED PROJECT PACKAGES

Object storage shall support versioning where available.

14. ARTIFACT VERSION PRESERVATION

An artifact associated with a finalized revision must not be silently overwritten.

The storage model shall preserve:

project_id
revision_id
model_version_id
artifact_type
artifact_version
content_hash
storage_reference
created_at

This permits recovery and provenance verification.

15. CONTENT HASHING

Recoverable artifacts shall be associated with cryptographic content hashes.

Conceptually:

ARTIFACT
↓
HASH
↓

STORAGE
↓
MANIFEST

Upon recovery:

RESTORED ARTIFACT
↓
RECALCULATE HASH
↓
COMPARE
↓
ACCEPT / REJECT

A hash mismatch shall prevent an artifact from being treated as verified.

16. PROJECT RECOVERY PACKAGE

OneDraft shall eventually support creation of a project recovery package containing sufficient metadata to reconstruct a project.

Conceptually:

PROJECT METADATA
MODEL STATE
REVISION HISTORY
CODE MANIFEST
VALIDATION RESULTS
DRAWING REFERENCES
DOCUMENT REFERENCES
ARTIFACT MANIFEST
AUDIT REFERENCES

The recovery package shall itself have a content hash.

17. RECOVERY MANIFEST

Every complete project recovery package shall contain a manifest identifying:

project_id
organization_id
current_revision_id
current_model_version_id
code_manifest_id
validation_run_id
artifact_references
artifact_hashes
export_timestamp
schema_version
application_version

This creates a deterministic recovery boundary.

18. APPLICATION BACKUP

Application services shall be treated as reproducible infrastructure rather than irreplaceable state.

The recovery source shall include:

SOURCE CODE
CONTAINER DEFINITIONS
PACKAGE DEFINITIONS
DATABASE MIGRATIONS
OPENAPI CONTRACT
DOMAIN TYPES
CONFIGURATION SCHEMAS
DEPLOYMENT DEFINITIONS

The application must be capable of being rebuilt from controlled source artifacts.

19. INFRASTRUCTURE RECOVERY

Infrastructure shall be represented through declarative configuration wherever practical.

The recovery system shall preserve:

CONTAINER CONFIGURATION
NETWORK DEFINITIONS
DATABASE CONFIGURATION
QUEUE CONFIGURATION
OBJECT STORAGE CONFIGURATION
SERVICE DEFINITIONS
RESOURCE POLICIES

Infrastructure configuration shall not exist solely as undocumented manual changes.

20. WORKER RECOVERY

Workers are disposable execution resources.

If a worker fails during a job:

JOB REMAINS RECOVERABLE
WORKER IS REPLACED
JOB IS REQUEUED OR FAILED SAFELY
PROJECT STATE REMAINS CONSISTENT

Workers must not become the sole storage location for project state.

21. QUEUE RECOVERY

The job execution system shall protect queued work against worker failure.

Jobs shall contain sufficient persistent metadata to reconstruct execution.

At minimum:

```
job_id
project_id
job_type
request_payload
model_version_id
revision_id
status
attempt_count
created_at
```

22. JOB IDEMPOTENCY

Recovery may result in a job being submitted more than once.

Therefore job execution shall support idempotency.

A recovered worker must determine whether:

```
JOB HAS NOT EXECUTED
JOB IS CURRENTLY EXECUTING
JOB COMPLETED
JOB FAILED
JOB WAS SUPERSEDED
```

before applying irreversible effects.

23. STALE JOB PROTECTION

A recovered job referencing an obsolete project state shall not modify the current model without revalidation.

For example:

```
JOB CREATED FOR REVISION 17
↓
PROJECT ADVANCES TO REVISION 18
↓
SYSTEM FAILURE
↓
```

JOB RECOVERED

The job must recognize that Revision 17 is no longer current.

It must either:

REBASE
REGENERATE
CANCEL

according to the operation type.

24. CONFIGURATION RECOVERY

Configuration required for production operation shall be recoverable.

Configuration includes:

SERVICE SETTINGS
DATABASE CONNECTION SETTINGS
QUEUE SETTINGS
STORAGE SETTINGS
MODEL CONFIGURATION
FEATURE FLAGS
JURISDICTION CONFIGURATION
DRAWING TEMPLATES
SYSTEM POLICIES

Secrets themselves shall be handled through the dedicated secrets-management mechanism.

25. SECRET RECOVERY

Secrets shall not be stored inside ordinary database backups.

The recovery architecture shall provide controlled recovery of:

DATABASE CREDENTIALS
API CREDENTIALS
ENCRYPTION KEYS
TOKEN SIGNING KEYS
OBJECT STORAGE CREDENTIALS
AI PROVIDER CREDENTIALS
THIRD-PARTY INTEGRATION CREDENTIALS

Secret recovery must require appropriate administrative authorization.

26. IDENTITY RECOVERY

Identity and authorization state must be recoverable independently of application services.

The recovery process shall preserve:

USERS
ORGANIZATIONS
MEMBERSHIPS
ROLES
AUTHORIZATION STATE
ACCOUNT STATUS

A restored project must not become accessible to an unauthorized organization because of incomplete identity recovery.

27. SECURITY INCIDENT RECOVERY

A security incident requires a different procedure from ordinary infrastructure failure.

The incident workflow shall be:

DETECT
↓
CONTAIN
↓
PRESERVE EVIDENCE
↓
REVOKE COMPROMISED CREDENTIALS
↓
ISOLATE AFFECTED SYSTEMS
↓
IDENTIFY LAST TRUSTED STATE
↓
RESTORE
↓
ROTATE SECRETS
↓
VALIDATE
↓
RESUME

28. TRUSTED RECOVERY POINT

The system must be able to identify the last known trusted recovery point.

A recovery point shall identify:

DATABASE TIMESTAMP
APPLICATION VERSION

SCHEMA VERSION
CONFIGURATION VERSION
ARTIFACT MANIFEST
SECURITY STATE

Recovery shall not blindly restore to the most recent backup if that backup may contain compromised state.

29. RPO

Recovery Point Objective defines the maximum acceptable amount of transactional data that may be lost.

The initial OneDraft production target shall be:

RPO TARGET: ≤ 15 MINUTES

The architecture should support tighter RPO as infrastructure maturity increases.

30. RTO

Recovery Time Objective defines the target time to restore service.

The initial target shall distinguish:

CRITICAL PLATFORM SERVICES
TARGET RTO: ≤ 4 HOURS

NON-CRITICAL DERIVED SERVICES
TARGET RTO: ≤ 12 HOURS

Production capacity may justify tighter objectives in later versions.

31. SERVICE PRIORITY

Recovery priority shall be:

1. DATABASE
2. IDENTITY
3. API
4. PROJECT MODEL SERVICES
5. QUEUE
6. WORKERS
7. OBJECT STORAGE ACCESS
8. VALIDATION
9. DOCUMENTATION GENERATION

- 10. FRONTEND
- 11. NON-CRITICAL INTEGRATIONS

The priority order may be adjusted according to incident type.

32. RECOVERY DEPENDENCY GRAPH

The recovery sequence shall respect service dependencies.

```
INFRASTRUCTURE
↓
DATABASE
↓
IDENTITY
↓
STORAGE
↓
API
↓
DOMAIN SERVICES
↓
QUEUE
↓
WORKERS
↓
VALIDATION
↓
DOCUMENTATION
↓
FRONTEND
↓
EXTERNAL INTEGRATIONS
```

33. DATABASE RESTORE PROCEDURE

The database restore procedure shall conceptually execute:

```
STOP MUTATIONS
↓
IDENTIFY TARGET RECOVERY POINT
↓
RESTORE BASE BACKUP
↓
APPLY WAL
↓
VERIFY DATABASE INTEGRITY
↓
VERIFY SCHEMA VERSION
↓
VERIFY CRITICAL TABLES
↓
```

VERIFY PROJECT REFERENCES
↓
OPEN READ-ONLY VALIDATION
↓
PROMOTE

34. DATABASE INTEGRITY VALIDATION

After restoration the system shall verify:

PRIMARY KEY INTEGRITY
FOREIGN KEY INTEGRITY
UNIQUE CONSTRAINTS
REVISION CHAINS
MODEL VERSION CHAINS
PROJECT REFERENCES
AUDIT EVENT SEQUENCES
CODE MANIFEST REFERENCES
DOCUMENT PACKAGE REFERENCES

35. PROJECT STATE VALIDATION

Each recovered project must be checked for consistency.

At minimum:

CURRENT REVISION EXISTS
CURRENT MODEL VERSION EXISTS
MODEL VERSION BELONGS TO PROJECT
REVISION BELONGS TO PROJECT
CODE MANIFEST IS ACCESSIBLE
REFERENCED ARTIFACTS EXIST
ARTIFACT HASHES MATCH

36. REVISION CHAIN VALIDATION

The revision graph shall be traversable from the current revision back to the project's initial revision.

Conceptually:

REVISION 1
↓
REVISION 2
↓
REVISION 3
↓
...

↓
CURRENT REVISION

Broken parent relationships shall be treated as a recovery integrity failure.

37. MODEL VERSION VALIDATION

Model versions shall be verified against their stored state hashes.

A mismatch indicates that the restored model state does not correspond to its recorded version.

Such a state shall not be promoted without investigation.

38. ARTIFACT MANIFEST VALIDATION

The artifact manifest shall be compared against object storage.

The recovery system shall detect:

MISSING ARTIFACT
DUPLICATE ARTIFACT
HASH MISMATCH
UNEXPECTED ARTIFACT
BROKEN REFERENCE

39. DOCUMENT PACKAGE VALIDATION

A document package shall identify:

PROJECT
REVISION
MODEL VERSION
CODE MANIFEST
VALIDATION RUN
ARTIFACT HASH

Recovery shall verify that all referenced components remain available.

40. COMPLIANCE STATE RECOVERY

Compliance data shall be recovered as versioned project state.

The system must preserve:

CODE SOURCES
CODE RULES
CODE MANIFESTS
RETRIEVAL DATES
SOURCE HASHES
APPLICABILITY
VALIDATION RESULTS

A recovered project must not silently acquire a different regulatory basis.

41. VALIDATION STATE RECOVERY

Validation runs are historical records.

They shall not be silently regenerated and substituted for the original result.

The system shall preserve:

validation_run_id
project_id
revision_id
model_version_id
code_manifest_id
results
timestamps
status

A regenerated validation may be recorded as a new validation run.

42. AUDIT EVENT RECOVERY

Audit events are append-only.

Recovery shall preserve their ordering and timestamps.

Recovered audit history must distinguish:

ORIGINAL EVENT
RECOVERY EVENT
RESTORATION EVENT
POST-RECOVERY EVENT

Recovery itself becomes an auditable event.

43. RECOVERY EVENT

The system shall create a recovery event containing:

recovery_id
incident_id
recovery_point
database_backup_reference
application_version
schema_version
operator_id
start_time
completion_time
result

44. BACKUP VERIFICATION

A backup shall not be considered reliable merely because it completed successfully.

The system shall verify:

BACKUP EXISTS
BACKUP IS READABLE
BACKUP HASH MATCHES
BACKUP METADATA IS VALID
RESTORE CAN BEGIN

Periodic complete restoration tests shall provide stronger assurance.

45. RESTORE TESTING

At defined intervals OneDraft shall execute restoration exercises.

The test shall:

SELECT BACKUP
↓
RESTORE TO ISOLATED ENVIRONMENT
↓
START SERVICES
↓
VERIFY DATABASE
↓
VERIFY PROJECTS
↓
VERIFY ARTIFACTS
↓
RUN INTEGRATION TESTS
↓
VERIFY DOCUMENT RECONSTRUCTION

46. RECOVERY DRILL

A full recovery drill shall test the ability to reconstruct the platform without relying on the original production instance.

The drill shall measure:

RECOVERY TIME
DATA LOSS
MISSING DEPENDENCIES
FAILED SERVICES
MANUAL STEPS
OPERATOR ERRORS
VALIDATION FAILURES

47. RECOVERY SCORE

Each recovery exercise shall produce a recovery assessment.

Conceptually:

RECOVERY SUCCESS
DATA INTEGRITY
SERVICE AVAILABILITY
ARTIFACT INTEGRITY
PROVENANCE INTEGRITY
SECURITY INTEGRITY
RPO PERFORMANCE
RTO PERFORMANCE

48. BUSINESS CONTINUITY MODES

OneDraft shall support operational states:

NORMAL
DEGRADED
READ_ONLY
RECOVERY
MAINTENANCE
EMERGENCY

49. READ-ONLY MODE

When transactional integrity cannot be guaranteed, the platform may enter:

READ_ONLY

In this mode:

PROJECT RETRIEVAL
DOCUMENT RETRIEVAL
HISTORICAL REVIEW
AUDIT REVIEW

may continue.

Mutating operations shall be disabled.

50. DEGRADED MODE

A degraded system may continue operating while non-critical services are unavailable.

For example:

DOCUMENT GENERATION OFFLINE

while:

PROJECT ACCESS
MODEL REVIEW
REVISION HISTORY

remain available.

51. EMERGENCY MODE

Emergency mode shall restrict the system to the minimum services required to protect project state and preserve operational continuity.

Potentially:

AUTHENTICATION
PROJECT READ
AUDIT
RECOVERY ADMINISTRATION

All other operations may be suspended.

52. REGIONAL FAILURE

The architecture shall eventually support recovery from loss of the primary infrastructure region.

Recovery resources should be capable of being provisioned independently.

The target architecture is:

```
PRIMARY REGION
  ↓
REPLICATED BACKUP
  ↓
SECONDARY RECOVERY REGION
```

53. STORAGE FAILURE

If primary object storage becomes unavailable, the system shall be able to restore artifacts from a secondary backup or archival source.

The project model must continue identifying the original artifact references and hashes.

54. DATABASE FAILURE

Database failure shall not cause loss of generated artifacts that are already stored independently.

Likewise, artifact storage failure shall not destroy the transactional project history.

This separation is fundamental to the recovery architecture.

55. CORRUPTION RECOVERY

Data corruption requires identification of the corruption boundary.

The system shall determine:

```
WHEN CORRUPTION BEGAN
WHICH PROJECTS ARE AFFECTED
WHICH REVISIONS ARE AFFECTED
WHICH ARTIFACTS ARE AFFECTED
WHICH BACKUP IS TRUSTED
```

Recovery shall use the latest trusted state rather than merely the latest timestamp.

56. PARTIAL PROJECT RECOVERY

OneDraft shall eventually support restoration of an individual project without restoring the entire platform.

The recovered project must preserve:

PROJECT MODEL
REVISION HISTORY
COMPLIANCE STATE
VALIDATION STATE
DRAWINGS
DOCUMENTS
AUDIT HISTORY

57. ORGANIZATION RECOVERY

The system shall eventually support organization-level recovery.

This includes:

ORGANIZATION
USERS
MEMBERSHIPS
PROJECTS
BILLING REFERENCES
ENTITLEMENTS
AUDIT EVENTS

Access controls must be reconstructed before project access is reopened.

58. RECOVERY ISOLATION

Recovered environments shall initially be isolated from production.

Conceptually:

BACKUP
↓
ISOLATED RECOVERY ENVIRONMENT
↓
VALIDATION
↓
SECURITY CHECK
↓
PROMOTION

No recovered environment should automatically become authoritative.

59. PROMOTION CRITERIA

A recovered environment may be promoted only after:

DATABASE VALIDATION PASSES
ARTIFACT VALIDATION PASSES
IDENTITY VALIDATION PASSES
APPLICATION HEALTH PASSES
SECURITY CHECK PASSES
INTEGRATION TEST PASSES

60. RECOVERY LOCK

During major recovery operations, mutating project operations shall be controlled through a recovery lock.

Conceptually:

RECOVERY_LOCK = TRUE

The API shall reject incompatible mutations until recovery validation is complete.

61. RECOVERY CONSISTENCY

The recovery process must preserve atomic relationships between:

MODEL
REVISION
GEOMETRY
VALIDATION
DRAWING
DOCUMENT
ACCEPTANCE

A recovered project must never expose a document generated from one revision while identifying another revision as its source.

62. VERSION COMPATIBILITY

Recovery must verify compatibility between:

DATABASE SCHEMA VERSION
APPLICATION VERSION
MODEL VERSION
ARTIFACT VERSION

OPENAPI VERSION
CONFIGURATION VERSION

Migration or compatibility procedures must execute before restored data is exposed to newer services.

63. DATABASE MIGRATION RECOVERY

Database migrations must be immutable and versioned.

The recovery process shall identify:

LAST APPLIED MIGRATION
TARGET MIGRATION
MIGRATION COMPATIBILITY

A migration must not be partially applied.

64. APPLICATION ROLLBACK

If a deployment causes operational failure, OneDraft shall support rollback to a previously known application version.

Rollback shall not automatically roll back database state unless an explicitly supported database migration reversal exists.

65. FORWARD-ONLY DATABASE MIGRATIONS

The preferred database strategy shall use forward-compatible migrations.

Conceptually:

VERSION N
↓
VERSION N+1
↓
VERSION N+2

Data preservation is preferred over destructive migration.

66. BACKWARD COMPATIBILITY

During deployment transitions, services may temporarily operate across adjacent application versions. The API and database contracts should therefore support controlled compatibility during deployment.

67. RECOVERY CONFIGURATION VERSION

Every recovery point shall identify the configuration version active at that time.

This prevents the system from restoring data under incompatible configuration assumptions.

68. TEMPLATE RECOVERY

Documentation templates, title blocks and graphic standards shall be treated as versioned recoverable assets.

A drawing generated using a specific template version must remain associated with that version.

69. JURISDICTION CONFIGURATION RECOVERY

Jurisdiction-specific configuration shall be recoverable independently.

This includes:

JURISDICTION IDENTIFIERS
CODE SOURCES
RULE CONFIGURATION
UNIT SETTINGS
DOCUMENT STANDARDS
LOCAL REQUIREMENTS

70. AI CONFIGURATION RECOVERY

AI orchestration configuration shall preserve:

MODEL PROVIDER

MODEL IDENTIFIER
PROMPT VERSION
TOOL POLICY
COMMAND POLICY
CONTEXT POLICY
EVALUATION CONFIGURATION

The objective is reproducibility of AI-assisted operations where technically possible.

71. AI OPERATION PROVENANCE

AI-generated commands must remain associated with:

COMMAND_ID
PROJECT_ID
REVISION_ID
MODEL_VERSION_ID
AI_CONFIGURATION_VERSION
REQUEST
RESULT
VALIDATION

Recovery must preserve this provenance.

72. EXTERNAL INTEGRATION RECOVERY

External integrations shall be independently classified as:

CRITICAL
IMPORTANT
OPTIONAL

Failure of an optional integration must not prevent core project recovery.

73. EVENT BUS RECOVERY

Event-driven operations shall be recoverable through persistent event records where required.

Events that are necessary to reconstruct state must not exist solely in volatile message queues.

74. EVENT REPLAY

Where supported, the system may replay recoverable events.

Replay must be:

IDEMPOTENT
VERSION-AWARE
PROJECT-AWARE
AUDITABLE

Events must not be replayed against an incompatible model version.

75. DUPLICATE EVENT PROTECTION

Every recoverable event shall have a unique event identifier.

Consumers shall prevent duplicate application of state-changing events.

76. OBSERVABILITY RECOVERY

The observability system shall preserve sufficient operational telemetry to diagnose recovery events.

Important records include:

ERROR LOGS
SERVICE HEALTH
JOB HISTORY
DATABASE HEALTH
RECOVERY EVENTS
SECURITY EVENTS

77. RECOVERY MONITORING

During recovery, monitoring shall track:

DATABASE RESTORE
WAL REPLAY
STORAGE RECOVERY
SERVICE STARTUP
QUEUE RECOVERY
WORKER RECOVERY
PROJECT VALIDATION
ARTIFACT VALIDATION

78. RECOVERY ALERTING

Critical recovery failures shall generate operational alerts.

Examples:

BACKUP FAILURE
WAL ARCHIVE FAILURE
HASH MISMATCH
DATABASE CORRUPTION
RESTORE FAILURE
MISSING ARTIFACT
REVISION CHAIN FAILURE
SECURITY VALIDATION FAILURE

79. BACKUP FAILURE POLICY

A failed backup must not silently replace the last known valid backup.

The system shall retain the previous valid recovery point until a newer backup has been verified.

80. BACKUP STATUS

Backup records shall support states:

STARTED
COMPLETED
VERIFIED
FAILED
EXPIRED
RESTORED
CORRUPTED

81. BACKUP METADATA

Each backup should record:

backup_id
backup_type
source_system
created_at
completed_at
size
content_hash
schema_version
application_version

storage_reference
verification_status
retention_class

82. RECOVERY JOB MODEL

Recovery operations shall execute through the OneDraft job architecture where practical.

Recovery jobs shall contain:

recovery_job_id
recovery_id
operation
source_reference
target_reference
status
progress
started_at
completed_at

83. RECOVERY AUTHORIZATION

Recovery operations require elevated authorization.

The system shall distinguish:

NORMAL ADMINISTRATION
RECOVERY ADMINISTRATION
SECURITY ADMINISTRATION

No ordinary project-level permission shall permit platform-wide recovery.

84. RECOVERY AUDIT

Every recovery action shall generate an audit event.

The audit record shall identify:

WHO
WHAT
WHEN
WHY
FROM WHERE
TO WHERE
RESULT

85. RECOVERY DOCUMENTATION

The repository shall contain documented recovery procedures.

Initial structure:

```
/docs/recovery/  
  database.md  
  artifacts.md  
  application.md  
  infrastructure.md  
  security.md  
  full-platform.md
```

86. RECOVERY AUTOMATION

Manual recovery steps should be minimized.

The long-term objective is:

```
DETECT  
↓  
CLASSIFY  
↓  
PROVISION  
↓  
RESTORE  
↓  
VALIDATE  
↓  
PROMOTE
```

with human authorization at critical promotion boundaries.

87. AUTOMATED BACKUP VERIFICATION

The system should automatically perform:

```
BACKUP EXISTENCE CHECK  
HASH CHECK  
METADATA CHECK  
RESTORE TEST  
DATABASE HEALTH TEST
```

according to configured schedules.

88. AUTOMATED PROJECT RECONSTRUCTION TEST

At least one representative project should periodically be reconstructed from stored recovery material.

The system shall verify:

PROJECT LOADS
MODEL LOADS
REVISION CHAIN LOADS
GEOMETRY REFERENCES RESOLVE
CODE MANIFEST RESOLVES
VALIDATION RESULTS LOAD
DRAWINGS RESOLVE
DOCUMENT PACKAGE RESOLVES

89. DOCUMENT REGENERATION TEST

A recovered project should eventually be capable of regenerating its associated documentation.

The test is:

RECOVER PROJECT
↓
LOAD MODEL VERSION
↓
LOAD REVISION
↓
LOAD CODE MANIFEST
↓
RUN VALIDATION
↓
GENERATE DRAWINGS
↓
GENERATE DOCUMENT PACKAGE

The resulting artifacts should be compared against the recorded provenance state.

90. RECOVERY ACCEPTANCE CRITERIA

A recovery is successful only when:

DATABASE IS CONSISTENT
PROJECTS ARE ACCESSIBLE
REVISION HISTORY IS INTACT
ARTIFACTS ARE AVAILABLE
HASHES MATCH
AUTHORIZATION IS CORRECT
SERVICES ARE HEALTHY

AUDIT EVENTS ARE PRESERVED
CORE INTEGRATION TESTS PASS

91. BUSINESS CONTINUITY TEST

The platform shall periodically demonstrate that it can continue operating after loss of a non-critical service.

For example:

DOCUMENT GENERATION FAILURE

must not prevent:

PROJECT ACCESS
MODEL REVIEW
REDLINE CREATION
REVISION MANAGEMENT

92. DEPENDENCY FAILURE TEST

The system shall test failure of:

DATABASE
QUEUE
WORKER
OBJECT STORAGE
AI PROVIDER
EXTERNAL INTEGRATION

where safe and appropriate.

The expected result must be defined for each failure.

93. AI PROVIDER OUTAGE

If the underlying AI provider becomes unavailable, the OneDraft project model shall remain operational.

The system must preserve:

PROJECT STATE
REVISION STATE
DOCUMENTATION
VALIDATION
AUDIT

AI-dependent operations may enter:

DEGRADED
QUEUED
UNAVAILABLE

without corrupting project state.

94. GEOMETRY ENGINE OUTAGE

If the geometry engine becomes unavailable, existing validated project state must remain readable.

New geometry operations may be queued until the engine becomes available.

95. DOCUMENT ENGINE OUTAGE

If document generation fails, previously generated document packages shall remain retrievable.

A failed generation must not overwrite a previously valid package.

96. COMPLIANCE ENGINE OUTAGE

Existing code manifests and validation results shall remain available.

A new compliance validation may be deferred until the compliance engine is restored.

97. RECOVERY OF FINAL ACCEPTED PROJECT

A finalized project must be recoverable as a complete historical state.

The recovery target shall include:

ACCEPTED REVISION
MODEL VERSION
CODE MANIFEST
VALIDATION
DRAWING SET
DOCUMENT PACKAGE
ACCEPTANCE RECORD

98. IMMUTABLE FINAL ARTIFACTS

Once a document package has been accepted and finalized, its underlying artifact references and hashes shall be preserved.

Subsequent project changes create new revisions rather than mutating the accepted historical package.

99. RECOVERY OF HISTORICAL PROJECT STATE

The system must support restoration or reconstruction of historical project states.

Given:

PROJECT_ID
REVISION_ID
MODEL_VERSION_ID

the system shall be capable of reconstructing the corresponding project state, subject to retained dependencies.

100. RECOVERY PROVENANCE CHAIN

The final recovery chain is:

```
BACKUP
↓
DATABASE
↓
PROJECT
↓
MODEL VERSION
↓
REVISION
↓
CODE MANIFEST
↓
VALIDATION
↓
DRAWING
↓
DOCUMENT PACKAGE
↓
ACCEPTANCE
↓
AUDIT
```

Every link must remain attributable.

101. RECOVERY SECURITY BOUNDARY

Recovery mechanisms must not become an alternate path around OneDraft authorization.

Recovered systems shall still enforce:

IDENTITY
ORGANIZATION
PROJECT
ROLE
SCOPE

boundaries.

102. RECOVERY ENVIRONMENT

The production repository shall define distinct recovery environments:

DEVELOPMENT
TEST
STAGING
RECOVERY
PRODUCTION

Recovery testing shall not modify production data.

103. RECOVERY DATA ISOLATION

Restored data used for testing shall be isolated from production identities and credentials.

Where appropriate, test environments shall use controlled synthetic or sanitized data.

104. BUSINESS CONTINUITY CONTACTS

Operational deployment shall eventually maintain defined responsibility for:

SYSTEM ADMINISTRATION
DATABASE ADMINISTRATION
SECURITY
APPLICATION OPERATIONS
INFRASTRUCTURE
RECOVERY AUTHORIZATION

These responsibilities shall be represented operationally rather than embedded as assumptions inside the software.

105. INCIDENT CLASSIFICATION

Recovery incidents shall initially be classified as:

INFRASTRUCTURE FAILURE
DATABASE FAILURE
STORAGE FAILURE
APPLICATION FAILURE
SECURITY INCIDENT
DATA CORRUPTION
REGIONAL OUTAGE
EXTERNAL DEPENDENCY FAILURE

106. INCIDENT SEVERITY

Initial severity classes:

SEV-1 – PLATFORM UNAVAILABLE OR DATA INTEGRITY THREAT
SEV-2 – MAJOR SERVICE DEGRADATION
SEV-3 – LIMITED SERVICE IMPACT
SEV-4 – NON-CRITICAL OPERATIONAL ISSUE

107. INCIDENT STATE

Recovery incidents shall progress through:

OPEN
INVESTIGATING
CONTAINED
RECOVERING
VALIDATING
RESOLVED
CLOSED

108. POST-RECOVERY REVIEW

Every significant recovery event shall produce a post-recovery review.

The review shall identify:

CAUSE
IMPACT
DATA LOSS
RECOVERY TIME
FAILED CONTROLS
SUCCESSFUL CONTROLS
CORRECTIVE ACTIONS

109. RECOVERY IMPROVEMENT LOOP

The recovery system shall continuously improve through:

INCIDENT
↓
RECOVERY
↓
ANALYSIS
↓
CORRECTIVE ACTION
↓
ARCHITECTURE UPDATE
↓
RECOVERY TEST

110. RECOVERY REGRESSION TESTING

Changes to:

DATABASE
STORAGE
MIGRATIONS
JOB SYSTEM
ARTIFACT SYSTEM
IDENTITY
SECURITY
INFRASTRUCTURE

shall trigger appropriate recovery regression testing.

111. RECOVERY DEPENDENCY INVENTORY

The platform shall maintain an inventory of dependencies required for complete recovery.

At minimum:

POSTGRESQL
OBJECT STORAGE
QUEUE
WORKERS
CONTAINER REGISTRY
SECRETS
IDENTITY
DNS
NETWORKING
AI PROVIDER
DOCUMENTATION ENGINE
GEOMETRY ENGINE
COMPLIANCE ENGINE

112. RECOVERY READINESS CHECK

Before a production release, the system should verify:

BACKUPS CURRENT
BACKUPS VERIFIED
WAL ARCHIVING ACTIVE
OBJECT STORAGE PROTECTED
MIGRATIONS AVAILABLE
CONTAINER IMAGES AVAILABLE
CONFIGURATION AVAILABLE
SECRETS RECOVERABLE
RESTORE PROCEDURES CURRENT
RECOVERY TEST STATUS ACCEPTABLE

113. RELEASE RECOVERY GATE

A production release shall not be considered complete unless the corresponding recovery capability remains available.

A release must not create an unrecoverable state.

114. BACKUP MONITORING DASHBOARD

The administrative control plane shall eventually expose:

LAST SUCCESSFUL BACKUP
LAST VERIFIED BACKUP
WAL ARCHIVE STATUS
ARTIFACT BACKUP STATUS
LAST RESTORE TEST
RECOVERY READINESS
CURRENT RPO

115. RECOVERY METRICS

Operational metrics shall include:

backup_success_total
backup_failure_total
backup_duration
backup_size
restore_duration
restore_success_total
restore_failure_total
artifact_integrity_failures
recovery_incidents
recovery_duration
rpo_actual
rto_actual

116. RECOVERY ALERT THRESHOLDS

Alerts should trigger when:

BACKUP AGE EXCEEDS POLICY
WAL ARCHIVING STOPS
BACKUP VERIFICATION FAILS
RESTORE TEST FAILS
ARTIFACT HASH VERIFICATION FAILS
RECOVERY TARGETS EXCEED RPO/RT0

117. ARCHIVAL STORAGE

Long-term project archives may be moved to lower-cost storage tiers while retaining their integrity and retrievability.

Archived state must remain associated with:

PROJECT
REVISION
MODEL VERSION
ARTIFACT HASH
RETENTION POLICY

118. ARCHIVAL RESTORATION

Archived projects shall be restorable into an isolated environment.

Restoration shall verify artifact integrity before exposing the project to normal services.

119. RETENTION POLICY INTEGRATION

Retention rules shall integrate with the OneDraft Data Lifecycle & Provenance System.

Backup retention must not independently delete information required by active project retention policies.

120. LEGAL AND REGULATORY HOLD

Where a project or artifact is subject to a preservation requirement, normal retention expiration must not delete the protected material.

The recovery architecture shall respect preservation flags established by the data lifecycle subsystem.

121. DELETION SAFETY

Deletion of source project data shall not automatically destroy required disaster-recovery material until the applicable retention policy permits removal.

Deletion operations shall be auditable.

122. RECOVERY PACKAGE EXPORT

The system shall eventually support a controlled export of a complete project recovery package.

Conceptually:

```
EXPORT PROJECT
↓
COLLECT MODEL
↓
COLLECT REVISIONS
↓
COLLECT COMPLIANCE
↓
```

COLLECT VALIDATION
↓
COLLECT DOCUMENTS
↓
COLLECT ARTIFACT MANIFEST
↓
HASH PACKAGE
↓
STORE

123. RECOVERY PACKAGE IMPORT

A recovery package may eventually be imported into a compatible OneDraft environment.

Import shall execute:

VERIFY SIGNATURE / HASH
↓
VERIFY VERSION COMPATIBILITY
↓
VERIFY REQUIRED ARTIFACTS
↓
VALIDATE PROJECT GRAPH
↓
IMPORT
↓
RUN RECOVERY TESTS

124. CROSS-VERSION RECOVERY

Recovery from an older OneDraft release into a newer release shall use controlled migration.

The system shall not assume that arbitrary historical versions are directly compatible.

125. SCHEMA EVOLUTION

Database schema evolution shall preserve the ability to read historical project state wherever technically feasible.

Destructive schema changes require explicit migration and archival planning.

126. RECOVERY TEST DATASET

The repository shall maintain a canonical recovery test project containing:

ORGANIZATION
BUILDING
MULTIPLE LEVELS
SPACES
WALLS
DOORS
WINDOWS
ASSEMBLIES
CONSTRAINTS
REVISION HISTORY
REDLINES
CODE MANIFEST
VALIDATION RESULTS
DRAWINGS
DOCUMENT PACKAGE
ACCEPTANCE

This project becomes the standard recovery fixture.

127. FULL PLATFORM RECOVERY TEST

The full test shall execute:

1. Destroy isolated application environment
2. Restore infrastructure
3. Restore PostgreSQL
4. Restore object storage
5. Restore configuration
6. Restore secrets
7. Restore identity
8. Start API
9. Start runtime orchestrator
10. Restore queues
11. Start workers
12. Restore project
13. Validate model
14. Validate revisions

15. Validate artifacts
 16. Validate compliance
 17. Validate documents
 18. Execute integration tests
 19. Confirm authorization
 20. Confirm audit continuity
-

128. RECOVERY ACCEPTANCE TEST

The recovery fixture shall pass:

PROJECT LOAD
MODEL LOAD
REVISION LOAD
GEOMETRY RESOLUTION
CODE MANIFEST RESOLUTION
VALIDATION RESOLUTION
DRAWING RESOLUTION
DOCUMENT RETRIEVAL
DOCUMENT REGENERATION
ACCEPTANCE RETRIEVAL
AUDIT RETRIEVAL

129. RECOVERY FAILURE CRITERIA

Recovery fails if:

PROJECT STATE IS INCONSISTENT
REVISION CHAIN IS BROKEN
MODEL HASH DOES NOT MATCH
REQUIRED ARTIFACT IS MISSING
ARTIFACT HASH DOES NOT MATCH
AUTHORIZATION IS INCORRECT
AUDIT HISTORY IS CORRUPTED
DOCUMENT PROVENANCE IS BROKEN
CORE SERVICES CANNOT START

130. BUSINESS CONTINUITY SUCCESS CRITERIA

The subsystem is successful when OneDraft can withstand:

DATABASE FAILURE
WORKER FAILURE
QUEUE FAILURE
STORAGE FAILURE
APPLICATION FAILURE
AI PROVIDER FAILURE
REGIONAL INFRASTRUCTURE FAILURE
SECURITY INCIDENT
DATA CORRUPTION

without losing the ability to reconstruct authoritative project state.

131. IMPLEMENTATION REPOSITORY

The recovery subsystem shall eventually contain:

```
/recovery
  /database
  /artifacts
  /infrastructure
  /configuration
  /security
  /procedures
  /tests
```

132. INITIAL IMPLEMENTATION ARTIFACTS

The first implementation package shall include:

```
backup configuration
database backup scripts
WAL archive configuration
artifact backup configuration
restore scripts
recovery validation scripts
recovery test fixture
recovery runbook
```

133. RECOVERY RUNBOOK

The production runbook shall provide deterministic procedures for:

DATABASE FAILURE
OBJECT STORAGE FAILURE
APPLICATION FAILURE
REGIONAL FAILURE
SECURITY INCIDENT
DATA CORRUPTION
FULL PLATFORM RECOVERY

134. AUTOMATION BOUNDARY

Automation may execute:

BACKUP
VERIFICATION
RESTORE PREPARATION
HEALTH CHECKS
INTEGRITY CHECKS
RECOVERY TESTS

Critical production promotion shall require explicit authorization.

135. HUMAN CONTROL BOUNDARY

The following actions require controlled authorization:

PRODUCTION RESTORE
RECOVERY PROMOTION
SECURITY INCIDENT RESTORE
DATA DELETION
RETENTION OVERRIDE
SECRET ROTATION
PRODUCTION FAILOVER

136. RECOVERY PROVENANCE

Every recovery operation must remain distinguishable from ordinary project activity.

The system shall never represent recovered data as though it were newly created data.

137. RECOVERY AND AUDIT SEPARATION

Recovery may restore historical audit records, but recovery operations themselves create new audit events.

Therefore:

HISTORICAL AUDIT
+
RECOVERY AUDIT

remain distinguishable.

138. RECOVERY AND CUSTOMER ACCEPTANCE

Customer acceptance records shall remain associated with the original accepted revision.

Recovery shall not create a new acceptance event merely because the platform was restored.

139. RECOVERY AND GENERATED DOCUMENTS

Previously generated documents remain historical artifacts.

A regenerated document is a new generation event and must not silently replace the original artifact.

140. RECOVERY AND DRAWING PROVENANCE

Recovered drawings shall preserve:

DRAWING_ID
REVISION_ID
MODEL_VERSION_ID
VIEW_ID
TEMPLATE_VERSION
ARTIFACT_HASH

141. RECOVERY AND COMPLIANCE PROVENANCE

Recovered compliance manifests shall preserve the original source versions and retrieval timestamps.

A current code source must not be substituted for the historical source used by a prior validation without explicit creation of a new manifest.

142. RECOVERY AND MODEL PROVENANCE

The project model remains the authoritative computational source.

Derived artifacts are recoverable consequences of the model.

Therefore the recovery system prioritizes:

MODEL
↓
REVISION
↓
VALIDATION
↓
DOCUMENTATION

rather than treating PDFs as the sole project record.

143. RECOVERY RECONSTRUCTION PRINCIPLE

If an artifact is lost but the authoritative model and sufficient generation metadata remain available, the artifact should be regenerated.

If the authoritative model is lost, artifact recovery alone is insufficient to reconstruct the full computational project state.

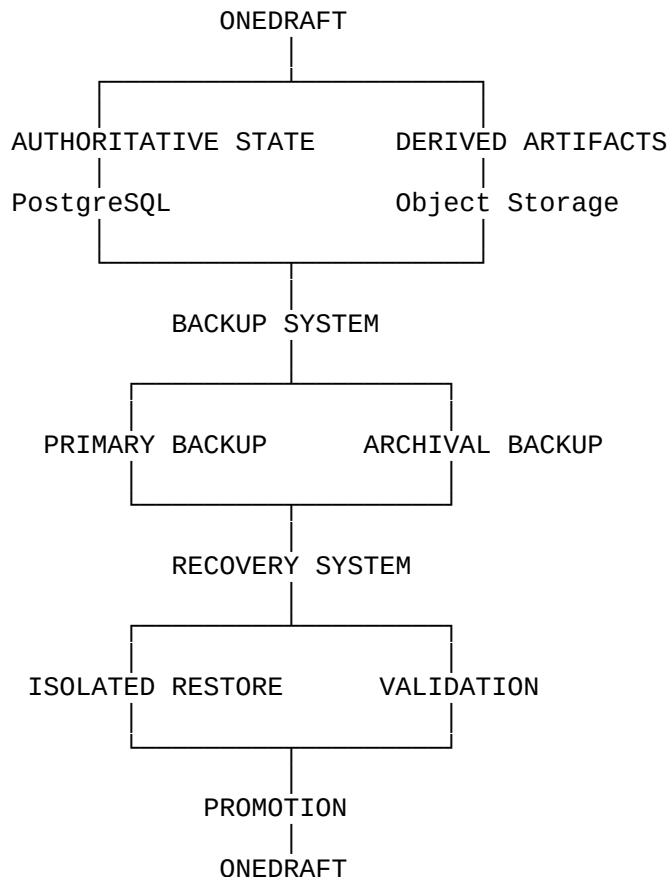
144. DISASTER RECOVERY ARCHITECTURAL PRINCIPLE

OneDraft shall be designed so that:

STATE IS DURABLE
WORK IS RESTARTABLE
ARTIFACTS ARE VERIFIABLE
SERVICES ARE REPRODUCIBLE
EVENTS ARE AUDITABLE
RECOVERY IS TESTABLE

145. FINAL RECOVERY ARCHITECTURE

The resulting architecture is:



146. END-TO-END RECOVERY CHAIN

The complete recovery chain becomes:

PROJECT
↓
MODEL
↓
REVISION
↓
COMPLIANCE MANIFEST
↓
VALIDATION
↓
DRAWING
↓
DOCUMENT
↓
ACCEPTANCE
↓
AUDIT
↓
BACKUP
↓
RESTORE
↓
RECONSTRUCTION

147. 35.0 ENGINEERING STATE

The OneDraft architecture now defines:

Backup architecture

Database recovery

Point-in-time recovery

Artifact recovery

Recovery manifests

Content verification

Project reconstruction

Application recovery

Worker recovery

Queue recovery

Configuration recovery

Secret recovery

Identity recovery

Security incident recovery

RPO

RTO

Business continuity modes

Recovery isolation

Recovery validation

Disaster recovery testing

Operational recovery

Historical state reconstruction

Recovery provenance

148. NEXT IMPLEMENTATION ARTIFACTS

The executable implementation package for 35.0 shall eventually include:

```
backup_manager.py  
database_backup.py  
artifact_backup.py  
restore_manager.py  
recovery_validator.py  
recovery_manifest.py  
recovery_jobs.py  
recovery_tests.py
```

and associated infrastructure configuration.

149. INTEGRATION WITH PRECEDING STACKS

The Backup, Disaster Recovery & Business Continuity subsystem integrates directly with:

```
Project Model  
Geometry Engine  
Compliance Engine  
Validation Engine  
Documentation Engine  
Review & Acceptance Engine
```

Runtime Orchestrator
Job & Worker Execution System
Artifact & Storage Management Engine
Integration & Event Bus
Identity & Security System
Observability System
Testing System
CI/CD System
Infrastructure System
Configuration System

No preceding subsystem is redefined.

150. FINAL ARCHITECTURAL DECISION

OneDraft shall not treat backup as a periodic copy of a database.

Backup and disaster recovery shall be implemented as a **system-wide state preservation and reconstruction capability**.

The platform must preserve sufficient information to answer:

WHAT WAS THE PROJECT?
WHAT MODEL VERSION EXISTED?
WHAT REVISION WAS CURRENT?
WHAT CODE MANIFEST APPLIED?
WHAT VALIDATION OCCURRED?
WHAT DRAWINGS WERE GENERATED?
WHAT DOCUMENTS WERE PRODUCED?
WHAT WAS ACCEPTED?
WHAT EVENTS OCCURRED?
CAN THE STATE BE RECONSTRUCTED?

The answer to each question must remain recoverable.

151. SPECIFICATION STATUS

ONEDRAFT BACKUP, DISASTER RECOVERY & BUSINESS CONTINUITY v0.1

STATUS: ESTABLISHED

The OneDraft architecture now contains the defined persistence, application, computational, compliance, validation, documentation, review, orchestration, execution, storage, event, security, observability, testing, deployment, configuration and recovery foundations.